Miguel Ángel Arroyo

Data & AI Engineer

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PROFESSIONAL PROFILE

Data & AI Engineer specialized in building pipelines that integrate heterogeneous data sources and apply LLMs to classify and structure information, using Python, Apache Airflow, PostgreSQL/PostGIS and local models (Ollama). Hands-on experience with RAG systems and with fine-tuning generative models, backed by quantitative evaluation of results.

PROFESSIONAL EXPERIENCE

InferIA · Alicante, Spain

Software Engineer · *Full-time, hybrid* | Oct 2026 – Present

- Designing the storage layer for heterogeneous information that will power an AI conversational assistant on WhatsApp (RAG architecture).

Data & AI Engineer · *Internship, remote* | Jun 2026 – Sep 2026

- Ingested public data sources in heterogeneous formats (XML, JSON, CSV): business registries, patent and trademark offices, scientific repositories and public procurement portals.
- Orchestrated the full workflow with **Apache Airflow** on Docker Compose on a GPU server: DAGs with task dependencies, scheduled runs and model concurrency control through pools.
- Built scrapers with requests and BeautifulSoup to extract the profiles of **400+ research groups** from two universities and their projects, with on-disk caching, exponential-backoff retries and robots.txt compliance.
- Classified and extracted structured information with **local LLMs (Ollama / Llama 3.1)**, normalized against a controlled catalog.
- Batch processing with resumable caches and idempotent loads into **PostgreSQL/PostGIS**, with version-controlled migrations (Alembic). Data source feasibility studies and technical documentation.

Technologies: Python, Apache Airflow, Docker, PostgreSQL/PostGIS, Alembic, Ollama/Llama 3.1, pandas, requests/BeautifulSoup, SQLite, Git.

FEATURED PROJECTS

Bachelor's Thesis — Fine-tuning pipeline for generative image models | Jan 2026 – Jul 2026

- Reproducible pipeline to adapt a generative model to a specific visual style: dataset creation and curation, labeling, training, inference and evaluation, across 4 incremental iterations.
- Trained **LoRA adapters on FLUX.1-dev** in the cloud (Modal, NVIDIA H100 GPU) and automated labeling with the **Florence-2** multimodal model.
- Quantitative evaluation with a custom script (**FID, CLIP score, DINOv2**): automatic labeling matched manual labeling, validating it as a way to scale the process.

Technologies: Python, PyTorch, AI Toolkit, torchao, FLUX.1-dev, LoRA, Florence-2, Modal.

m.ai — Personal notes assistant on WhatsApp | Jun 2026

- Bot with a full **RAG architecture**: notes by text or voice, natural-language semantic search and summaries. Audio transcription with Whisper, image understanding, LLM-based auto-tagging and a vector database with recency and relevance re-ranking.

Technologies: Python, FastAPI, Groq API (gpt-oss-20b, Whisper, qwen3.6-27b), Upstash Vector, Meta Cloud API, Render.

DevRadar — Tech job market demand analysis | In progress

- ELT pipeline that ingests job postings from the Adzuna API, classifies technologies with an LLM, transforms the data with **dbt on DuckDB** and visualizes it in a Streamlit dashboard.

Technologies: Python, dbt-core, DuckDB, Groq / Ollama, GitHub Actions (weekly cron), Streamlit.

TECHNICAL SKILLS

AI: Local LLMs (Ollama), RAG, agents, vector databases, fine-tuning (LoRA), dataset creation and labeling, model evaluation, prompt engineering, AI-assisted development (Claude Code, Google Antigravity).

Data: Python, SQL, pandas, ETL/ELT pipelines, Apache Airflow, dbt, DuckDB, PostgreSQL/PostGIS, Alembic, web scraping, API and public data source integration.

Development: Docker / Docker Compose, Git, GitHub Actions, FastAPI, Streamlit, React, Angular, Node.js, MongoDB.

EDUCATION AND CERTIFICATIONS

B.Eng. in Multimedia Engineering · University of Alicante | 2022 – 2026

Certifications: Databricks Fundamentals Accreditation (Databricks, Sep 2026) · Claude Code 101 (Anthropic, Aug 2026)

LANGUAGES

Spanish: Native. **English:** Fluent technical reading / Intermediate.